

1 Mitochondria supply energy for the processes in cells.

(a) (i) How many of these statements about the inner membrane of a mitochondrion are correct?

(1)

- it is folded (has cristae)
- it is the location of ATP synthase
- it is the site of the electron transfer chain
- it is a site of protein synthesis

☐ **A** 1

☐ **B** 2

☐ **C** 3

☐ **D** 4

(ii) Anaerobic respiration produces lactate as a waste product.

Which is the result of lactate production?

(1)

- ☐ **A** blood becomes less acidic
- ☐ **B** muscle becomes less fatigued
- ☐ **C** pH of the blood increases
- ☐ **D** muscle becomes more acidic

(b) Describe how chemiosmosis is involved in the synthesis of ATP.

(2)

(c) A culture containing respiring cells was supplied with radioactive oxygen molecules.

Which will be the first substance to contain radioactive oxygen atoms?

(1)

- ☐ **A** ATP
- ☐ **B** carbon dioxide
- ☐ **C** reduced NAD
- ☐ **D** water

(Total for Question 1 = 5 marks)